

Edmilson Roque dos Santos

Research interests: Dynamical Systems, Ergodic Theory, Synchronization, Sparse Recovery methods, Reservoir Computing.

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🌐 <https://github.com/edmilson-roque-santos>

PROFESSIONAL APPOINTMENTS

Postdoctoral Researcher

Sep 2025 – to date

NONLINEAR DYNAMICS AND TIME SERIES ANALYSIS

MAX PLANCK INSTITUTE FOR THE PHYSICS OF COMPLEX SYSTEMS

Project: *Learning sparse network dynamics from data*

Research Associate

Feb 2023 – May, 2025

CLARKSON CENTER FOR COMPLEX SYSTEMS SCIENCE

CLARKSON UNIVERSITY

NSF-NIH CRCNS: *Functional Brain Networks with Tensioned Stability for Optimal Processing*

EDUCATION

PhD. in Applied Mathematics

Feb 2018 – Jan 2024

ICMC-USP AND IMPERIAL COLLEGE LONDON (PARTIALLY SUPPORTED BY ROYAL SOCIETY).

Project Title: *Reconstruction of Complex Networks from Data*

Supervisor: Prof. Dr. Tiago Pereira

Co-Supervisor: Prof. Dr. Sebastian van Strien

MSc. in Physics

Feb. 2016 – Feb. 2018

IFSC-USP

Thesis title: *Discontinuous transitions to collective dynamics in star motifs of coupled oscillators*

Supervisor: Prof. Dr. Tiago Pereira

Co-Supervisor: Dr. Jaap Eldering

Bachelor in Physics

Feb. 2012 – Dec. 2015

IFSC-USP

Undergraduate research title: *Models in Explosive Synchronization*

Supervisor: Prof. Dr. Francisco Aparecido Rodrigues

PUBLICATIONS

PREPRINTS AND IN PREPARATION

1. Edmilson Roque dos Santos. “*Learning sparse network dynamics from transient macroscopic measurements*”. In preparation.
2. Daniel Estevez Moya, Francesco Martinuzzi, Edmilson Roque dos Santos, Erick Alejandro Madrigal Solis, Ernesto Estevez Rams, and Holger Kantz. “*Characterizing the better performance of multi-model ensemble forecasts through reservoir computing*”. In preparation.
3. Edmilson Roque dos Santos. *Reconstruction of bursting network dynamics from data*. Under review.
4. Edmilson Roque dos Santos, Sebastian van Strien, and Tiago Pereira. “*Ergodic Basis Pursuit induces Divide-and-Conquer Network Reconstruction*”. In preparation.

JOURNAL PUBLICATIONS

1. Edmilson Roque dos Santos and Erik Bollt. “On the emergence of numerical instabilities in Next Generation Reservoir Computing”. <https://doi.org/10.1063/5.0278709>. Chaos 35, 123102 (2025).
2. Yuanzhao Zhang, Edmilson Roque dos Santos, Huixin Zhang, and Sean P. Cornelius. “How more data can hurt: Instability and regularization in next-generation reservoir computing”. <https://doi.org/10.1063/5.0262977>. Chaos 35, 073102 (2025).
3. Tiago Pereira, Edmilson Roque dos Santos, Sebastian van Strien. “Robust reconstruction of sparse network dynamics”. <https://iopscience.iop.org/article/10.1088/1361-6544/add3b0>, Nonlinearity 38 055031 (2025).
4. Anil Kumar, Edmilson Roque dos Santos, Paul J. Laurienti, and Erik Bollt. “Symmetry breaker governs synchrony patterns in neuronal inspired networks”. Chaos 34, 113115 (2024). <https://doi.org/10.1063/5.0209865>
5. Erik Bollt, Jeremie Fish, Anil Kumar, Edmilson Roque dos Santos, and Paul J. Laurienti. “Fractal Basins as a Mechanism for the Nimble Brain”. Sci Rep 13, 20860 (2023). <https://doi.org/10.1038/s41598-023-45664-5>.
6. Juliano Genari, Guilherme T. Goedert, Sérgio H.A. Lira, Krerley Oliveira, Adriano Barbosa, et al. “Quantifying protocols for safe school activities”. PLoS ONE 17(9): e0273425 (2022). <https://doi.org/10.1371/journal.pone.0273425>
7. Marcel Novaes, Edmilson Roque dos Santos, Tiago Pereira. “Recovering sparse networks: Basis adaptation and stability under extensions”. Physica D: Nonlinear Phenomena 424 132895, (2021). <https://doi.org/10.1016/j.physd.2021.132895>
8. Jaap Eldering, Jeroen Lamb, Tiago Pereira, Edmilson Roque dos Santos. “Chimera states through invariant manifold theory”. Nonlinearity 34-5344, (2021). <https://dx.doi.org/10.1088/1361-6544/ac0613>

GRANTS AND HONORS

2025	SIAM Travel Awards - Life Sciences and Dynamical Systems Travel Fund by Dr. Simone Bianco.
2018 - 2022	Doctoral Scholarship: The São Paulo Research Foundation, FAPESP.
2016 - 2018	Grant: Coordination for the Improvement of Higher Education Personnel, CAPES.
2015	Garfield’s Medal: Best Oral Presentation in the Symposium of Mathematics for the Undergraduate course (SiM 2015) at ICMC-USP.
July 2015	Research internship at PIK under the supervision of Prof. Jurgen Kurths - The São Paulo Research Foundation, FAPESP.
2014 - 2015	Undergraduate Scientific Initiation Scholarship: The São Paulo Research Foundation, FAPESP.

CONFERENCES AND INVITED TALKS

Apr 27, 2026	PSU-Purdue-UMD Joint Seminar on Mathematical Data Science. <i>Learning network dynamics from data through compressive sensing techniques</i> . (Invited talk)
Apr 20 - 24, 2026	One-Dimensional Dynamics and High-Dimensional Network Systems, Palazzo del Castelletto, Pisa. <i>Learning network dynamics from data through compressive sensing techniques</i> . (Invited talk)
Mar 8 - 13, 2026	DPG meeting - Statistical Physics and Nonlinear Dynamics - Dresden. <i>Learning spatiotemporal patterns from mean-field data</i> . (Contributed talk)

Jan 12-13, 2026	International Workshop: Recent Trends in Coupled Network Systems January 12 - 13, 2026 — Berlin, WIAS. <i>Learning spatiotemporal patterns from mean-field data</i> . (Poster presentation)
Oct 9, 2025	Math Seminar at Center for Systems Biology Dresden. <i>Ergodic Basis Pursuit leads to exact reconstruction of sparse network dynamics</i> . (Invited talk)
May 11, 2025	MS41 - Invariant Sets in Dynamics: Applications and Future Directions. SIAM Conference on Applications of Dynamical Systems 2025. <i>Switching Between Multiple Invariant Sets in Neuronal-Inspired Network Dynamics</i> . (Invited talk)
Dec 18, 2024	(Network) Dynamical Systems. University of São Paulo, São Carlos. <i>Symmetry breaker governs synchrony patterns in neuronal inspired networks</i> . (Contributed talk)
Oct 7, 2024	Applied Math Seminars. University of Ottawa, Canada. <i>Metastability of chimeras states in coupled networks</i> . (Invited talk)
July 8 - 12, 2024	Fourth Symposium of Machine Learning on Dynamical Systems. Fields Institute, Toronto - Canada <i>Ergodic Basis Pursuit induces exact (and robust) sparse network reconstruction</i> . (Contributed talk)
Mar 20 - 22, 2024	NERCCS 2024: Seventh Northeast Regional Conference on Complex Systems. Potsdam, NY - USA. <i>Dynamics of synchrony patterns on networks</i> . (Contributed talk)
Feb 5, 2024	Oberseminar Dynamics. TUM, Munich - Germany. <i>Ergodic Basis Pursuit induces robust reconstruction of sparse network dynamics</i> . (invited talk - online format)
Oct 27, 2023	C3S2 Seminars. The Clarkson Center for Complex Systems Science, Potsdam, NY - USA. <i>Reconstruction of coupled sparse networks from data</i> . (invited talk)
Mar 22 - 24, 2023	NERCCS 2023: Sixth Northeast Regional Conference on Complex Systems. <i>Ergodic Basis Pursuit induces robust network reconstruction</i> . (contributed talk)
Jan 9 - 11, 2023	Dynamics Days US 23. <i>Ergodic basis pursuit induces robust reconstruction of weakly coupled sparse networks</i> . (contributed talk - online format)
Dec 17 - 18, 2022	Mathematical Physics Days 2022. <i>Reconstruction of Weakly Coupled Sparse Networks from Data</i> . (invited talk - online format)
Sep 12 - 21, 2022	Inverse Network Dynamics - NETDAT22. MPI for the Physics of Complex Systems, Dresden - Germany. <i>Ergodic basis pursuit induces robust reconstruction of sparse networks</i> . (contributed talk)
Jun 20 - 23, 2022	Rényi 100. Hungarian Academy of Sciences, Budapest - Hungary. <i>Ergodic basis pursuit induces robust reconstruction of sparse networks</i> . (poster presentation)
May 6, 2022	Free University of Berlin (FUB). Berlin - Germany. <i>Ergodic basis pursuit induces robust reconstruction of sparse networks</i> . (presentation)
May 5, 2022	Potsdam Institute for Climate Impact Research (PIK). Potsdam - Germany. <i>Ergodic basis pursuit induces robust reconstruction of sparse networks</i> . (presentation)
May 5, 2022	Weierstrass Institute for Applied Analysis and Stochastic (WIAS). Berlin - Germany. <i>Ergodic basis pursuit induces robust reconstruction of sparse networks</i> . (presentation)
May 4, 2022	University of Potsdam. Potsdam - Germany. <i>Ergodic basis pursuit induces robust reconstruction of sparse networks</i> . (presentation)
May 23 - 27, 2021	SIAM Conference on Applications of Dynamical Systems (DS21). <i>Ergodicity implies stable reconstruction of sparse network dynamics</i> . (invited talk - online format)
April 29, 2021	Dynamical Systems and Networks Seminars. Courant Institute of Mathematical Sciences, New York - USA. <i>Chimera states through invariant manifold theory</i> . (invited talk - online format)
Oct 07 - 11, 2019	V Escola Brasileira de Sistemas Dinâmicos. UFMG - Belo Horizonte, MG - Brazil. <i>Chimera states through invariant manifold theory</i> . (poster presentation)
Aug 26 - Sep 01, 2018	V Workshop and School on Dynamics, Transport and Control in Complex Networks - ComplexNet. INPE, Cachoeira Paulista, SP - Brazil.

Discontinuous transitions to collective dynamics in star motifs of coupled oscillators. (poster presentation)

- Jul 27 - 31, 2015 International Workshop on Dynamics of Coupled Oscillators: 40 years of the Kuramoto Model. MPI for the Physics of Complex Systems, Dresden - Germany.
Title: *Influence of frequency distribution on the discontinuous phase transition in networks of Kuramoto oscillators.* (poster presentation)
- Sep 17 - 19, 2014 Undergraduate Research Project Highlights 22^o SIICUSP -The University of São Paulo's International Symposium of Undergraduate Research (SIICUSP) São Paulo - Brazil.
Title: *Influence of frequency distribution on the discontinuous phase transition in networks of Kuramoto oscillators.* (poster presentation)
- Oct 6 - 11, 2014 III Workshop and School on Dynamics, Transport and Control in Complex Networks - ComplexNet. São José dos Campos, SP - Brazil.
Title: *Influence of frequency distribution on the discontinuous phase transition in networks of Kuramoto oscillators.* (contributed talk)

VISITING

- August 2025 University of São Paulo, ICMC-USP, Brazil
PROF. DR. TIAGO PEREIRA, DR. THOMAS PERON AND DR. EDDIE NIJHOLT
- Mar - Jun 2022 Imperial College London, London, UK.
PROF. DR. SEBASTIAN VAN STRIEN AND PROF. DR. JEROEN LAMB
- Dec 2019 - Mar 2020 Imperial College London, London, UK.
PROF. DR. SEBASTIAN VAN STRIEN AND PROF. DR. JEROEN LAMB
- Jan - May, 2019 Imperial College London, London, UK.
PROF. DR. SEBASTIAN VAN STRIEN AND PROF. DR. JEROEN LAMB
- Jan - Feb, 2018 Imperial College London, London, UK.
PROF. DR. JEROEN LAMB
- Jul - Aug, 2015 Potsdam Institute for Climate Impact Research - PIK, Potsdam, Germany
PROF. DR. JURGEN KURTHS

ORGANIZING

- Jul 19 - 24, 2026 Minisymposium at Dynamics Days - Europe 2026 on *Advances in Reconstruction of Network Dynamics from Time Series*
CO-ORGANIZATION WITH BENGI DÖNMEZ
- Jun 28 - Jul 3, 2026 Half a Century of the Kuramoto Model: Explaining Emergent Order in Time and Space
CO-ORGANIZATION WITH DR. SIMONA OLMÍ, PROF. DR. MICHAEL ROSENBLUM, AND PROF. DR. MARC TIMME
- Jan, 19-20, 2026 Mini workshop on Reservoir Computing for Nonlinear Dynamics
SMALL WORKSHOP AT MPI-PKS
- Oct, 2025 - ongoing Nonlinear time series analysis seminars at MPI-PKS
CO-ORGANIZATION WITH PROF. DR. HOLGER KANTZ
- Mar 20 - 22, 2024 NERCCS 2024
CO-ORGANIZATION WITH PROF. DR. CHUNLEI LIANG, PROF. DR. ERIK BOLLT, DR. GOLSHAN MADRAKI, AND DR. JEREMIE FISH
- 2022 - 2023 Dynamical systems research seminars at ICMC-USP
CO-ORGANIZATION WITH DR. ZHENG BIAN
- Jun - Jul, 2022 Dynamical systems research seminars
ORGANIZATION OF SEMINARS AT IMPERIAL COLLEGE LONDON

Apr - Dec, 2021	NetDynamics Seminars between ICMC-USP and Nodds Lab – Dr. Deniz Eroglu’s group from Kadir Has University (KHAS) CO-ORGANIZATION WITH DR. ELIF YUNT.
Oct 24 - 26, 2018	São Paulo Dynamical Systems days CO-ORGANIZATION WITH PROF. DR. TIAGO PEREIRA AND PROF. DR. ALI TAHZIBI
2014	SIFSC 4 - Semana Integrada de Física de São Carlos CO-ORGANIZATION WITH UNDERGRADUATE AND GRADUATE STUDENTS FROM IFSC-USP
2013	SIFSC 3 - Semana Integrada de Física de São Carlos CO-ORGANIZATION WITH UNDERGRADUATE AND GRADUATE STUDENTS FROM IFSC-USP

PEER REVIEW

Physical Review Letters; Physical Review Applied; Physical Review Research; Chaos: An Interdisciplinary Journal of Nonlinear Science; Physical Review E; Physica A: Statistical Mechanics and its Applications; Physica D: Nonlinear Phenomena; and Chaos, Solitons and Fractals;

PROGRAMMING LANGUAGES

Advanced *Python*
Basic *C++*, *Fortran*, *Matlab*, *Julia*, *Mathematica*

TEACHING

2019	TA for Linear Algebra and Ordinary Differential Equations UNIVERSITY OF SÃO PAULO Undergraduate Course
2017	TA for Advanced Laboratory of Physics UNIVERSITY OF SÃO PAULO Undergraduate Course
2017	TA for Mathematical Physics UNIVERSITY OF SÃO PAULO Undergraduate Course

EXTRA ACTIVITIES

2020 - 2022	Modcovid19. Participation in a large collaboration group formed by different Brazilian institutions to model COVID-19 in Brazil, in particular, model validation of COMORBUSS software, which can be accessed in the following link: https://comorbuss.org/Home .
2014	Judge during the finals in Brazil. INTERNATIONAL YOUNG PHYSICISTS TOURNAMENT (IYPT)